Chapter 1

Bivariate Stable Functions and Directional Depth

1.1 Bivariate Stable Functions and Directional Depth

In this chapter we study bivariate independent stable random variables and explore the directional depth function in a new light. Recall that we consider a symmetric stable random variable

$$X \sim S(\alpha, 0, 1, 0)$$

with characteristic function

$$\varphi_X(t) = \exp(-|t|^\alpha).$$

Let X and Y be independent copies. Then, for any fixed constants a, b, the linear combination

$$Z = aX + bY$$

has characteristic function

$$\varphi_Z(t) = \exp\Big[-\Big(|a|^\alpha + |b|^\alpha\Big)|t|^\alpha\Big],$$

implying that

$$Z \sim S\Big(\alpha, 0,\ c',\ 0\Big), \quad \text{with} \quad c' = \Big(|a|^\alpha + |b|^\alpha\Big)^{1/\alpha}.$$

This scaling property lets us standardize the problem so that

$$P\Big(aX + bY \geq au + bv\Big) = P\left(\frac{aX + bY}{c'} \geq \frac{au + bv}{c'}\right).$$

For brevity we set

$$c = \frac{au + bv}{c'},$$

which represents the standardized directional threshold.

In our discussion the direction (u, v) is always taken as a unit vector (with respect to the chosen norm). This is analogous to the treatment in [1].

1.1.1 Analysis of the Directional Threshold Function and its Optimization

We study the directional threshold function defined by

$$c(a,b) = \frac{a\,u + b\,v}{\left(|a|^\alpha + |b|^\alpha\right)^{1/\alpha}},$$

where the direction (u,v) is assumed to be a unit vector and the weights (a,b) are chosen under the normalization, since $c(a,b)$ is homogeneus of degree 1.

$$|a|^\alpha + |b|^\alpha = 1.$$

A convenient parameterization is

$$a = t^{1/\alpha}, \quad b = (1-t)^{1/\alpha}, \quad t \in [0,1],$$

so that the function becomes

$$c(t) = u\,t^{1/\alpha} + v\,(1-t)^{1/\alpha}.$$

Concavity versus Convexity. The nature of the mapping $x \mapsto x^{1/\alpha}$ is crucial:

- For $\alpha > 1$: Since $1/\alpha < 1$, the function $x^{1/\alpha}$ is *concave*. Thus, $c(t)$ is a concave function on $[0,1]$, and its maximum is attained at a unique interior point.
- For $0 < \alpha \leq 1$: Here, $1/\alpha \geq 1$ so that $x^{1/\alpha}$ is *convex*; hence, $c(t)$ is convex on $[0,1]$ and the maximum is attained at an endpoint.

Optimization for Different α Cases.

Case 1: $\alpha > 1$

Differentiating $c(t)$ and setting $c'(t) = 0$ leads quickly to

$$\left(\frac{1-t}{t}\right)^{\frac{\alpha-1}{\alpha}} = \frac{v}{u},$$

which yields the optimal value

$$t^* = \frac{1}{1 + \left(\frac{v}{u}\right)^{\frac{\alpha}{\alpha-1}}}.$$

Substituting t^* into $f(t)$ we obtain

$$c_{\max} = u\left[1 + \left(\frac{v}{u}\right)^{\frac{\alpha}{\alpha-1}}\right]^{\frac{\alpha-1}{\alpha}}.$$

Defining

$$\beta = \frac{\alpha}{\alpha-1},$$

this simplifies to the elegant form

$$\boxed{c_{\max} = \left(u^\beta + v^\beta\right)^{1/\beta}.}$$

Case 2: $0 < \alpha \leq 1$

When $0 < \alpha \leq 1$ the function $c(t)$ is convex, and therefore its maximum is attained at one of the endpoints of the interval. Hence, in this case we have

$$\boxed{c_{\max} = \max\{u, v\}.}$$

1.2 Integral Representation for the Depth Function

Once the threshold $s_{\max}$ is determined, we evaluate the tail probability of a standard symmetric stable variable as follows. For

$$X \sim S(\alpha, 0, 1, 0),$$

the Gil–Pelaez inversion formula states that the CDF is

$$F(x) = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \frac{\sin(tx)}{t}\exp(-t^\alpha)\,dt.$$

Thus, the tail probability becomes

$$P(X \geq x) = 1 - F(x) = \frac{1}{2} - \frac{1}{\pi}\int_0^\infty \frac{\sin(tx)}{t}\exp(-t^\alpha)\,dt.$$

In our bivariate framework, after standardizing the linear combination with the scale factor $c' = \left(|a|^\alpha + |b|^\alpha\right)^{1/\alpha}$ and using the directional threshold $au + bv$, the probability of interest is

$$P\Big(aX + bY \geq au + bv\Big) = P\Big(X \geq c\Big),$$

where

$$c = \frac{au + bv}{c'}.$$

Plugging in the optimal threshold $s_{\max}$ (which now depends on the relative values of u and v and on α) we obtain the final formula for the depth:

$$\boxed{\operatorname{depth}(u, v) = P\Big(aX + bY \geq au + bv\Big) = \frac{1}{2} - \frac{1}{\pi}\int_0^\infty \frac{\sin\Big(t\, s_{\max}\Big)}{t}\exp(-t^\alpha)\,dt.}$$

Here, note that:

- For $\alpha > 1$: $s_{\max} = \left(u^\beta + v^\beta\right)^{1/\beta}$ with $\beta = \frac{\alpha}{\alpha - 1}$.
- For $0 < \alpha < 1$: $s_{\max} = \max\{u, v\}$.

1.3 Special Cases and Depth Contours and Visualization

For illustration, consider two scenarios:

- When $0 < \alpha < 1$ (say, $\alpha = 0.8$), the function $f(t) = u\,t^{1/\alpha} + v\,(1-t)^{1/\alpha}$ is convex so that the maximum is attained at one of the endpoints. Hence, the optimal threshold is

$$s_{\max} = \max\{u, v\}.$$

- When $\alpha > 1$ (say, $\alpha = 1.3$), the optimum is given by

$$s_{\max} = \left(u^\beta + v^\beta\right)^{1/\beta}, \quad \beta = \frac{1.3}{1.3 - 1} \approx 4.33.$$

Figure 1.1 shows the depth contours for $\alpha = 0.8$ and Figure 1.2 illustrates the contours for $\alpha = 1.3$. Notice that for $\alpha > 1$ the contours are given by the L^β-norm and tend to be rounder, while for $\alpha < 1$ the depth depends solely on the larger coordinate, leading to more "degenerate" contour shapes.

These examples also "prove" that our formulation is consistent with classical cases: for $\alpha = 1$ (the Cauchy case) one obtains $s_{\max} = \max\{u, v\}$ and a tail probability of

$$P(X \geq x) = \frac{1}{2} - \frac{1}{\pi}\arctan(x),$$

and for $\alpha = 2$ (the Gaussian case) with $\beta = 2$, the threshold becomes

$$s_{\max} = \sqrt{u^2 + v^2},$$

which is the familiar Euclidean norm. These special cases along with the contour plots vividly illustrate the rich geometric behavior of the depth function as α varies.

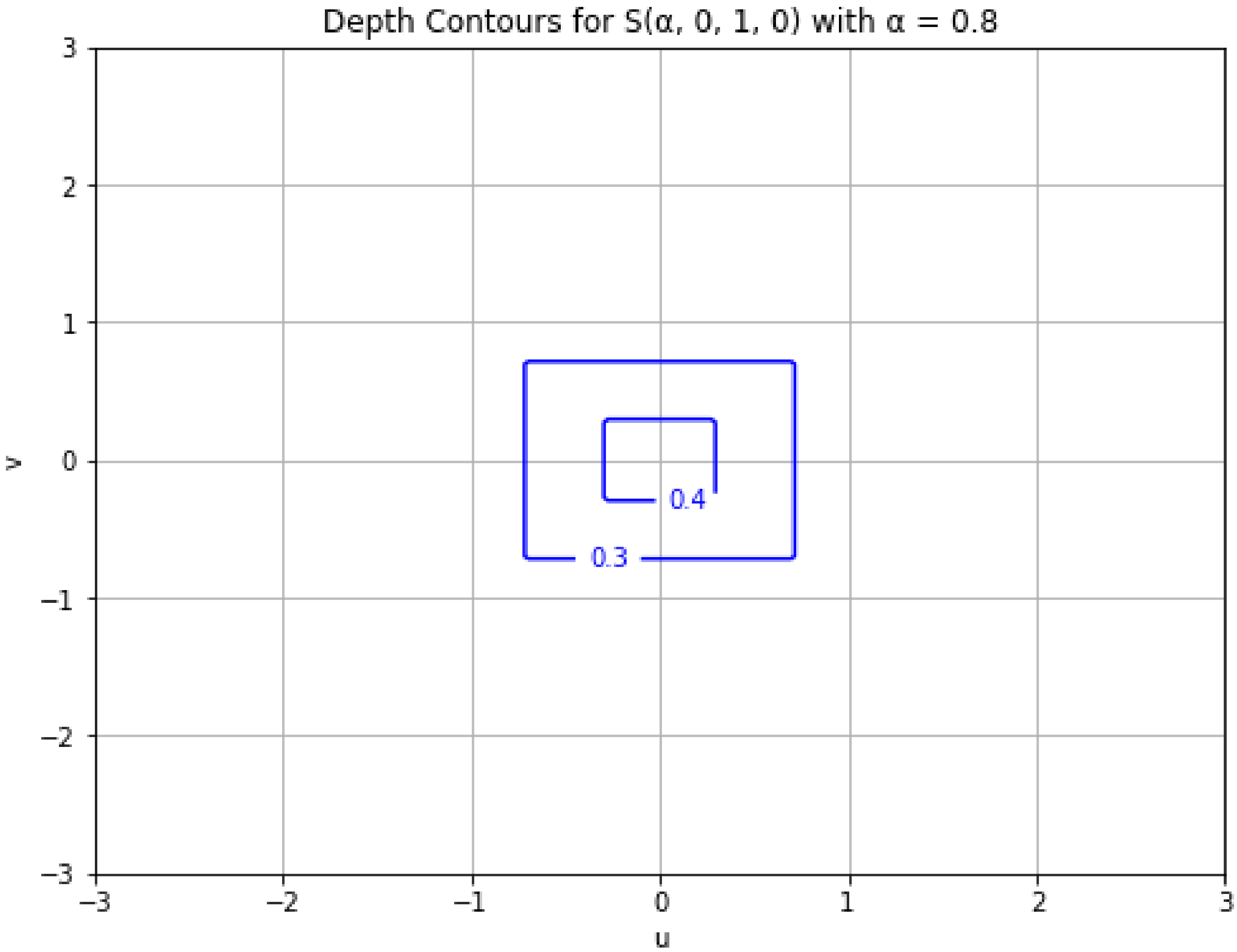

Figure 1.1: Depth contours for $\alpha = 0.8$.

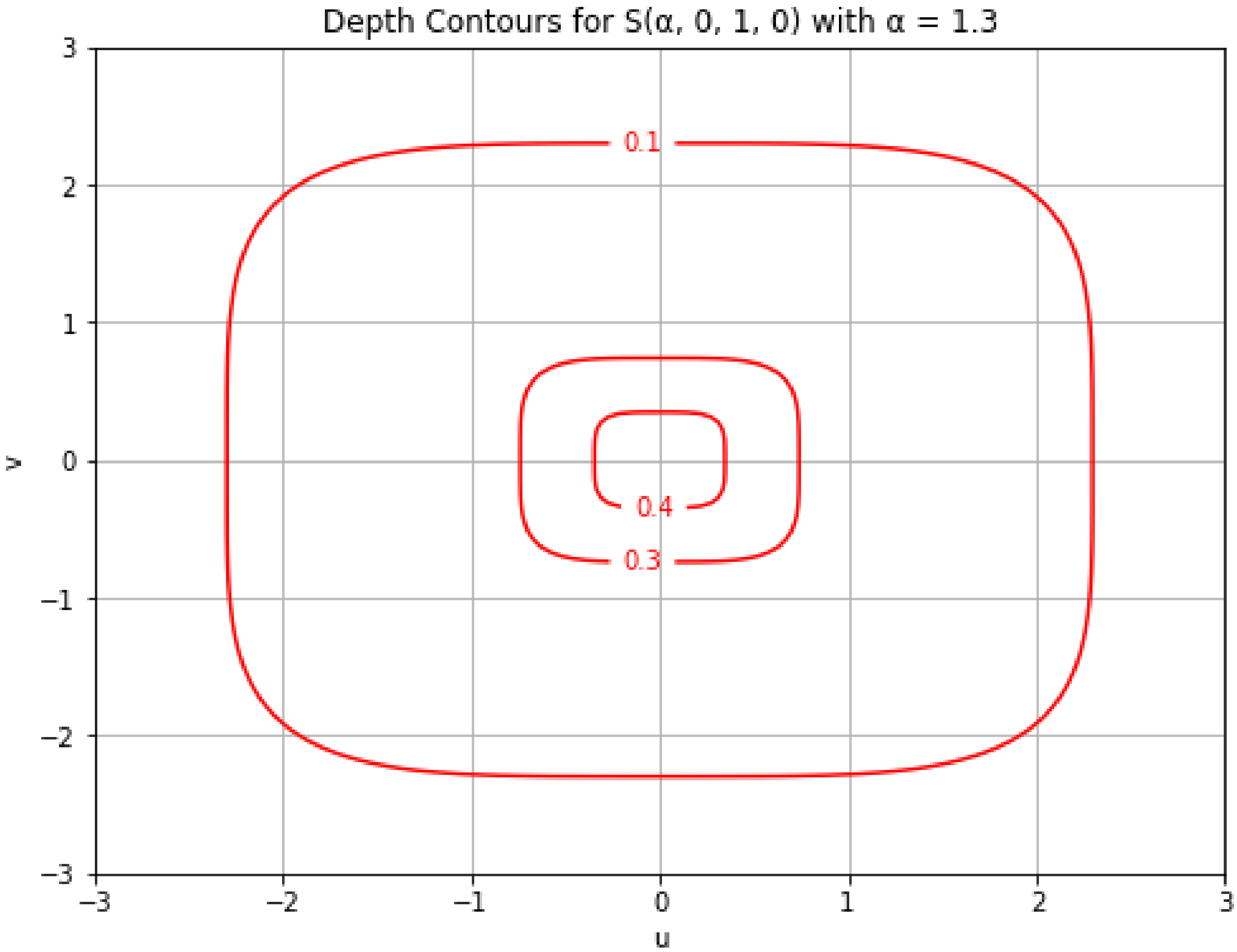

Figure 1.2: Depth contours for $\alpha = 1.3$.

Bibliography

[1] Rousseeuw, P. J., & Ruts, I. (1999). The depth function of a population distribution. *Metrika*, 235–236.